Say Less, Never Mislead

Cross-Layer Selective Abstention in an Offline Assistive Perception System, Extended to a Tool-Mediated Voice Agent

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ABSTRACT

A sighted person using an assistive app checks what it says almost without noticing, and when it is wrong they drop the claim and lose nothing. A blind person has no cheap way to run that check, so a confident wrong answer is accepted and acted on, and it costs more than silence would have. We built BlindAssist, an offline camera-based guidance system for blind and low-vision users, around that difference, and the design idea we want to argue for is that a system should be able to decline to answer at every layer where it can be wrong, with each layer deciding on its own terms rather than on one shared confidence score. We describe five such mechanisms, covering distance estimation, path advice, network failure, how often the system is allowed to interrupt, and finally a voice agent where a local offline language model chooses among fourteen deterministic capabilities and writes none of the guidance. On 200 labelled utterances, putting a keyword parser in front of the model raises overall routing accuracy from 39.5% to 53.0% and keeps 100% on the phrasings the parser already covers, which an LLM-only router drops to 45.0%, and the same model asked to answer in prose instead of choosing a tool invents perceptual content in 42.5% of its replies. The cost is that out-of-scope over-triggering rises from 5.0% to 55.0%. We then recorded two speakers reading the same utterances aloud, and most of what the agent layer gained disappears: its margin over the keyword baseline falls from 6.8 points to 0.9, while the deterministic layer's abstention barely moves. Both of those are negative results and we report them as they came out.

CCS Concepts: • Human-centered computing → Accessibility technologies; Ubiquitous and mobile computing; • Computing methodologies → Artificial intelligence.

Keywords: blind and low-vision users, assistive technology, abstention, on-device machine learning, voice agents, tool use, hallucination

1 INTRODUCTION

A sighted person using an assistive app is running a check the whole time without thinking about it. The app says "chair on your right", they glance right, there is no chair, and they drop the claim and carry on. Nothing in the design of the app has to account for that, because the check is free and it happens whether the designer planned it or not.

A blind user does not have that check. Every spoken claim is either accepted or acted on, because there is no cheap way to test it against anything, and this is not a hypothetical worry. Blind readers of machine-generated image captions have been observed building detailed and confident interpretations of descriptions that were simply wrong, since nothing in the interaction gave them a reason to doubt [24], and the same thing shows up again with generative assistants, where fluency itself gets read as a sign of reliability [1, 45]. The human-factors literature has the other half of it: automation that is unreliable does not just fail to help, it makes people stop using the aid at all [32, 20], and past a certain point an imperfect alert leaves the user worse off than no alert would have [49, 5].

So the default that most interactive systems are built on, which is to answer something rather than nothing, is the wrong default here. A confident wrong answer costs the user more than silence, because silence costs them one query and a wrong answer costs them their sense of when the system can be trusted at all, and that is much harder to get back.

This paper reports BlindAssist, a working offline guidance system with a Python reference implementation and an Android/Flutter port that share one pure-logic core and mirrored test suites. The claim we want to make about it is architectural:

Given that the user cannot verify what they are told, the ability to decline should be designed into every layer separately, with each layer deciding on criteria that come from its own failure mode, rather than being handed to one confidence threshold at the output.

We support that with five mechanisms already running in the system, an agent layer built on the same principle, and an evaluation that was frozen before the agent existed. The results include two findings that did not go the way we expected, and we have kept both.

2 RELATED WORK

Crowd- and cloud-backed visual question answering set up the interaction [4] and the dataset tradition [13] that this system inherits, and it did so against a background where a lot of everyday visual content simply is not available to a blind user at all [28]. The scene-description apps in use today [3, 42, 9, 23] give rich descriptions, but they depend on latency and connectivity, they are verbose by design, and they are not built for the continuous business of not walking into things. Dedicated wearables [31] read near-field text very well at the price of buying hardware. Indoor navigation assistants locate the traveller rather than the objects in front of them [2, 39, 11], spatial-audio wayfinding works at beacon granularity [44], and electronic travel aids [48] report proximity without saying what the object is or exactly where; surveys [19] and comparative studies [53] cover the space. Work on blind users training their own recognisers [17] starts from the same premise we do, that getting the recogniser right is not the whole problem. What we could not find in this literature was a system where declining to answer is treated as a behaviour to be designed and measured rather than as the error path.

Abstention itself is old. Classification with a reject option goes back to Chow [6], and it has a modern treatment in selective prediction [8, 10], learning to defer [25, 29] and confidence estimation [14, 12]. All of that asks when a model should abstain and answers with a threshold on a score. We are asking a different question, about an interface rather than a classifier, and our answer is that five separate layers of one system each have their own way of being wrong and should each abstain on grounds that come from that.

Fluent generation that is not supported by anything is well described [15, 26], and a model's own confidence signal only tells you so much [18]. Tool use [40, 52, 33, 35] and constrained decoding [41, 50] are ordinary engineering by now. What we are adding is the reason for applying them: when the person on the other end cannot look and see that the answer is wrong, containment stops being a quality measure and becomes a safety one, and the target changes from making fabricated perception rare to making it something the system cannot express. We report the ablation that shows what that costs.

Obstacle sonification [27, 7] and vibrotactile direction [46] are established, so the modalities are not our contribution. What we did was drive all of them from one ordinal localisation core, so speech, stereo sonar and haptics carry the same meaning and share the same timing rules.

3 SYSTEM

A handset streams camera frames to a tethered laptop over Wi-Fi, the laptop runs YOLOv8s [38, 16] trained on COCO [21] together with a small custom door and dustbin detector, and everything the user actually hears or feels stays on the phone: speech, stereo-panned proximity sonar, haptics, grammar-constrained offline speech recognition [47, 34] and on-device OCR. All the guidance logic sits in a layer with no camera, no model and no clock of its own, and that layer is written twice, once in Python and once in Dart, with test suites that mirror each other (221 and 159 tests). Figure 1 shows how it fits together.

Objects are located ordinally, by which we mean a 3×3 zone grid taken from the box centre and a per-class proximity bucket of very close, close, medium or far taken from box area. A distance in metres does exist, but it is gated (§4).

Running the detectors on the phone measured about 2.5 s per frame for both the GPU and the NNAPI delegate, which is why inference is tethered at all. On the laptop GPU the two detectors together cost 21 ms, while total server time per frame during a field walk was about 305 ms, so almost all of the remaining cost is reconstructing and moving the frame rather than looking at it.

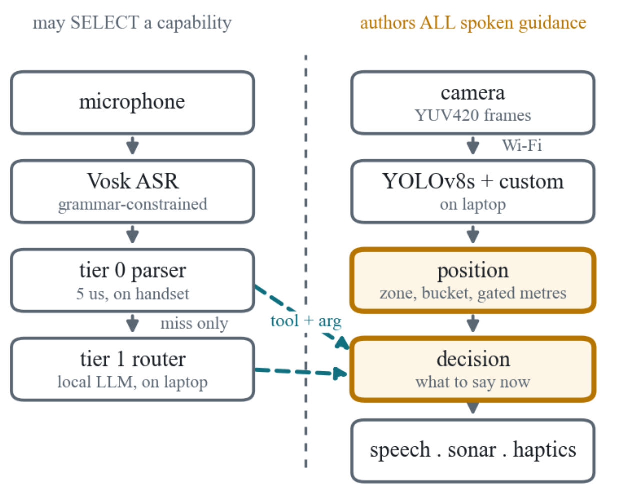

Figure 1. The two dashed arrows crossing the boundary carry the whole safety argument. The router is allowed to point at a capability and is not allowed to speak through one. Tier 0 runs on the handset, so every trained phrase still works with the laptop switched off.

4 FIVE WAYS TO DECLINE

4.1 *Perception: distance that is gated on reliability*

Distance comes out of a pinhole model, $d = h_{real} \cdot F / h_{box}$, and it is spoken as "about N metres" only when three conditions hold, which are that the box does not touch a frame edge, that the class name clears a confidence threshold, and that the object is not already close. The first of those matters most. A box cut off by the edge of the frame is shorter than the object really is, so the model reads it as further away, and that error points the wrong way for exactly the nearest and largest things in the scene, which are the ones most likely to be clipped. When any of the three fails the system falls back to the ordinal bucket, which it can always say honestly. Learned monocular depth [37] would give a better estimate, but it would not remove the need for the gate, because its failure cases are quiet too.

4.2 *Planning: path advice with a threshold on openness*

Each of left, ahead and right is scored by how close its nearest obstacle is, rather than by how much area is occupied in that third. Scoring by area inverts in a case that comes up constantly, where a bulky object further away outranks a small hazard close by, and the advice then points the user at the hazard. Doorways are left out of the obstacle mass, since a door is a thing to walk through. If even the emptiest third has a close obstacle in it, the system does not pick the least bad direction, it says "Stop, no clear path", because a recommender that always names a direction has no way of saying that none of them is any good.

4.3 *Transport: absence is not the same as a negative*

A frame that fails to come back returns `null` and never an empty list. An empty list means the scene was looked at and found clear, and acting on that when it is not true has three specific effects: the sonar goes quiet, which means "path clear", the escalation state resets, and Find reports "not visible" for something that may be right in front of the user. On no data the engine pauses guidance and says that it has, rather than turning its own outage into a confident answer.

4.4 *Attention: fewer warnings, and a way to ask*

This one came out of the first field walk rather than out of the design. The verdict was that it kept saying all the objects and it was too much of a cluster, which is the alarm-fatigue result [43, 5] arriving in a setting where the channel being flooded is the only guidance channel the user has. A warning that is not heeded has negative value, since it spent attention and gave nothing back. Continuous warnings now fire only at close range or nearer, and everything taken out of that channel can still be asked for: `check` answers "is there anything in front of me?" from the same ordinal core, in tier 0, with no model and no network involved. A user who asks has by asking given the attention that the continuous channel is not allowed to assume.

4.5 *Dialogue: abstaining from routing*

An unknown tool, an unknown object class, a missing argument, prose where JSON should be, a timeout, or any exception at all become an abstention, and even the clarifying question that gets asked is picked from a fixed table by key rather than written by the model.

5 THE AGENT LAYER

BlindAssist's offline recogniser is grammar-constrained, so only phrases in an explicit list can be transcribed at all, and free speech is therefore not mis-parsed, it is never heard in the first place. A trigger word opens a dictation window, which is transcribed either by a local Whisper model [36] on the laptop or, with the laptop off, by a second recogniser built on the same 40 MB model that is already loaded with its grammar removed. The transcript goes to the deterministic parser first and only reaches the router if that misses (Figure 2).

The model is given the tool registry, the class enumeration, a deterministic state block listing the visible classes with zone, proximity and count along with the current mode, the last announcement and object memory, and the utterance, and it has to return JSON. Everything past that point treats what comes back as untrusted input. A reference like "is it still there" therefore gets resolved against what the detector saw and not against a description, so perception never goes back into the model.

We should say where the boundary moved, because it did. The system as first built enforced a stronger rule, that no spoken token whatsoever came from the model. After the first field walk the developer-user asked for conversation, by which he meant the ability to ask a question in his own words and get an answer instead of getting routed, and we granted that and nothing more. A reply travels in a separate channel that the executor cannot push through any capability, it is grounded in the same state block, it is length-capped and cut at a sentence, loose prose emitted where a tool call belongs is still thrown away, and one flag puts the absolute rule back for the ablation in §7.2. So the honest version of the claim is that no guidance token comes from the model, which is weaker than where we started, and we would rather say that than let the earlier wording stand.

On the handset the two tiers sit on opposite sides of a Wi-Fi link, and the client re-validates whatever the server sends against the same closed registry before executing any of it, so the containment guarantee does not depend on trusting the network. A reply with one unusable action in it is thrown away whole rather than partly executed, since running the half that happened to parse would itself be an unverified action.

Fourteen capabilities (Table 4) used to be declared in four places, a parser, its phrase list, a dispatcher and the Dart equivalents, and those places had already drifted apart: the web dispatcher had silently stopped handling five capabilities that the parser could still produce. One declarative table now drives the recogniser's phrase list at runtime, generates the model's tool schema, backs a single executor, and emits a committed manifest that both test suites check field by field. We claim consistency that is enforced across sites, not single-source generation, because the field-validated parser is left in place on purpose rather than being regenerated from the table, and rewriting it would risk more than the stronger claim is worth.

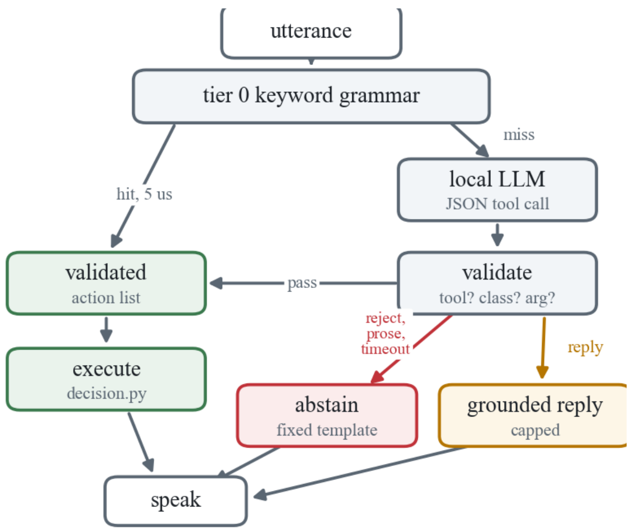

Figure 2. Two-tier routing. Every path except the amber reply channel produces text that was written by deterministic code or picked from a fixed template, and the reply channel cannot reach any capability's output.

6 EVALUATION

Before any of the routing work, the perception and guidance layers were checked on seven recorded phone clips, which between them produced 31 announcements. Every announcement saves an annotated keyframe and all 31 were reviewed against the image. Direction was right on 31 of 31 and no announcement was made for something that was not there, while 6 of the 31 used a wrong class name and still gave the correct warning, which is the pattern behind saying the generic word “obstacle” when confidence is below threshold. These are small numbers reviewed by the author, so we report them as a sanity check on the deterministic core and not as a perception result.

The protocol and a set of 200 labelled utterances were frozen and committed before the router was written, so that the router could not be tuned against its own test set. The categories are canonical (40, phrasings the grammar covers), paraphrase (70), multi-intent (20), out-of-scope (40, where the gold label is to abstain) and ambiguous (30, where the gold answer depends on a state block stored in the record). The out-of-scope records are plausible things somebody would say to an assistive device, so weather, calls, messages, time and battery, rather than nonsense strings, because nonsense is easy to abstain on and would have made the metric look better than it is.

There are three configurations, keyword-only, LLM-only with tier 0 stubbed to always miss, and two-tier. Routing accuracy is exact match on the ordered action list, and a two-action utterance routed to one correct action scores zero, because the user asked for two things and got one. The over-trigger rate is the share of out-of-scope utterances that got answered with any action at all, and it is the safety metric for this layer. The fabrication check confirms that every spoken string that was executed is one the deterministic core could have produced for that record, or a fixed template, and the harness fails loudly on anything else. Intervals are Wilson [51], and we are not claiming significance between configurations at this set size, the comparison is descriptive.

Two speakers who did not author the set read a stratified 60-record subset aloud, once each, in a normal room. The recordings were transcribed by the same recogniser the handset uses for open dictation, with the grammar removed, so this is the condition the system actually runs in rather than a simulation of it. Utterance boundaries were recovered by aligning the recognised word stream to the known script, and where fewer than a third of an utterance's words aligned the record was dropped instead of being paired with a transcript we could not trust, which left 107 transcripts over 59 records. That gate takes out the utterances the recogniser handled worst, so the spoken numbers below are if anything a little kind to the system.

Unless stated otherwise the model is `llama3.2:3b` [22, 30] under Ollama on the tether laptop. What would falsify each claim was written into the protocol before any of the runs.

7 RESULTS

Category	n	keyword	LLM-only	two-tier
canonical	40	100.0	45.0	100.0
paraphrase	70	0.0	48.6	47.1
multi-intent	20	0.0	30.0	10.0
out-of-scope	40	95.0	47.5	45.0
ambiguous	30	3.3	43.3	43.3
overall	200	39.5	45.0	53.0

Table 1. Routing accuracy (%), clean text, n = 200. Wilson 95% CIs on the overall row: keyword 33.0–46.4, LLM-only 38.3–51.9, two-tier 46.1–59.8.

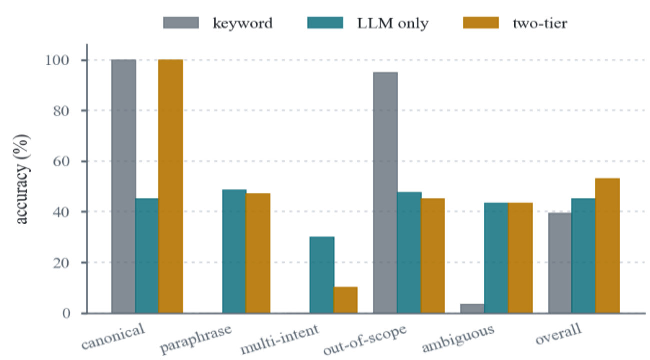

Figure 3. Routing accuracy by category, n = 200. For out-of-scope, accuracy means correctly abstaining. The keyword bars sitting at zero are not degradation, they are absence, since a grammar cannot hear what is not in its grammar.

The overall column in Table 1 is the least interesting part of it.

The shape of the keyword row is really the motivation for this work restated as data, since it is perfect on the phrasings it was built for and sits at zero on paraphrase and multi-intent, and zero there is absence rather than degradation.

Putting the parser in front of the model beats using the model alone, and it does so where we expected. LLM-only loses 55% of the canonical category, because the model mis-routes utterances the parser gets right by construction, usually by picking a plausible neighbour such as `describe` for `count` or `walk` for `zones`. Two-tier holds 100% there because those utterances never reach the model at all, and it still picks up most of the paraphrase gain, so the deterministic tier is both faster and more accurate on the traffic it covers.

The part we did not expect is what happens to abstention. The keyword baseline abstains on 95% of out-of-scope input and gets that for nothing, because an utterance it cannot match returns nothing at all, and both language-model configurations spend almost all of it (Table 2). A 3B model choosing among fourteen tools will nearly always find one it likes, so "call my mum" becomes `read`, "what time is it" becomes `clock`, and "take a photo" becomes `walk`. We are reporting that unmitigated, because the protocol was frozen before the router existed and tuning the prompt against this set is the thing the freeze exists to prevent. It does not undo the containment result, since no run fabricated perception and tier-0 abstention is untouched for the traffic tier 0 covers, but it does rule out any reading in which adding a local model is free.

	keyword	LLM-only	two-tier
abstained	38	19	18
answered anyway	2	21	22
over-trigger rate	5.0%	52.5%	55.0%
tier-0 hit, p50	5 µs	—	5 µs
tier-1 hit, p50	—	1172 ms	1188 ms
tier-1 hit, p95	—	1578 ms	1500 ms
served by tier 0	30.0%	0%	30.0%
conversational replies	0	4	6

Table 2. Out-of-scope behaviour (n = 40) and routing latency. Wilson 95% CIs on the over-trigger rates: keyword 1.4–16.5, LLM-only 37.5–67.1, two-tier 39.8–69.3.

Both of the keyword over-triggers are substring collisions, since "read my email" contains *read* and "how do i get to the bus stop" contains *stop*. They are what matching on keywords costs, and they are exactly the sort of error a router with sentence-level context ought to clean up, except that it does not, because tier 0 claims them before the model is ever consulted. That is the other side of putting the parser first and we would rather state it than leave it out.

Per-call latency at tier 1 is much the same in both configurations, 1172 against 1188 ms at the median, so two-tier's advantage there is entirely about how much traffic reaches the model rather than how fast the model is. 30% of utterances are served at 5 µs, and that 30% is made up of the commands users issue most often. Tier 1 is usable for a question the user asked and unusable inside the continuous guidance loop, and every capability here happens to be the former, which is a fact about this system rather than a general result.

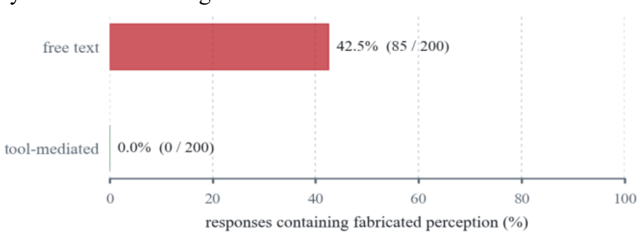

Figure 4. The same model with the same deterministic state block. Asked to choose a tool it fabricated nothing in any configuration; asked to answer, it fabricated in 42.5% of replies. The zero comes from construction rather than tuning, and the 42.5% is a keyword-based lower bound that only counts invented objects.

Figure 4 is the cleanest thing in the paper, and what the model invents is worse than the rate makes it sound. Asked to "find bottle" with a state block that listed no bottle, it replied:

> *"i'm walking in front of you, my cane tapping on the ground. i've stopped about 6 feet away from your right side. there's a small…"*

It has invented an object, a distance in feet, a bearing, and a first-person travelling companion carrying a cane. A sighted user would notice and dismiss it. The user this system is for gets it in the same voice, at the same volume, with the same confidence as a real detection.

7.1 What real speech does to all of this

	keyword text	keyword spoken	two-tier text	two-tier spoken
canonical	100.0	84.2	100.0	84.2
paraphrase	0.0	0.0	16.7	13.0
multi-intent	0.0	0.0	8.3	8.7
out-of-scope	91.7	91.3	41.7	30.4
ambiguous	0.0	0.0	58.3	52.6
overall	37.3	34.6	44.1	35.5
over-trigger	8.3	8.7	58.3	69.6

Table 3. Accuracy (%) on the matched subset, meaning the same 59 records scored first as written text and then as 107 real transcripts from two speakers. The subset's category mix is different from the full set, so these columns compare with each other and not with Table 1.

The spoken condition changes what we can claim, which is why the protocol asked for it in advance.

The deterministic tier barely reacts to it. Keyword accuracy goes from 37.3 to 34.6 and its over-trigger rate is flat, 8.3 against 8.7, because a grammar matching on a few content words still matches when the recogniser drops a function word, and "how much battery is left" came back as "how much batteries left" and still abstained.

Most of what the agent layer was buying goes away. Two-tier beats the baseline by 6.8 points on written text, 44.1 against 37.3, and by 0.9 points on the same utterances spoken, 35.5 against 34.6. The paraphrase coverage that the agent exists to provide is mostly destroyed before it gets there, because a paraphrase is long and full of unconstrained vocabulary and that is exactly what a 40 MB open-dictation model transcribes worst. An evaluation on clean text therefore overstates what this layer gives a user who speaks to it, and we would not have known that if the protocol had not asked for the condition.

The last one we did not anticipate at all. Under ASR the keyword configuration's out-of-scope abstention holds up, 91.7 to 91.3, while two-tier's falls from 41.7 to 30.4 and its over-triggering goes from 58.3% to 69.6%. A garbled out-of-scope utterance does not read to a language model as a reason to decline, it reads as more room to interpret. The layer that abstains because of how it is built keeps abstaining when the input gets worse, and the layer that abstains because it judged the input abstains less, at exactly the moment the input got harder to judge. We had argued for building abstention structurally and this is the first measurement we have that says so.

7.2 Model size, and putting the absolute rule back

Two follow-up runs on the same frozen set, both with the same model family and harness. The first was the ablation the design promised, where `allow_chat` is turned off so that no spoken token at all can come from the model, which is the rule the system started with. It scored 53.0% overall with a 55.0% over-trigger rate, identical to the shipped configuration on both. Conversational replies happened 6 times in 200 and a reply carries no action, so it already scored as an abstention; opening that channel bought the user something without costing the routing metrics anything, and it also did not fix anything.

The second asks whether over-triggering is about model size, which is the first thing a reader would want to rule out. Running the same two-tier configuration on `llama3.2:1b` gives 49.5% overall, lower than 3B as expected, and an over-trigger rate of 55.0%, which is exactly the 3B figure. Going smaller by a factor of nearly three changed how often the model routed correctly and did not change how often it refused to decline at all. We are not going to over-read two points on one axis, but the easy explanation, that a bigger model would be more willing to abstain, does not survive its first test.

8 DISCUSSION AND LIMITATIONS

The system has been field-walked by one sighted developer, so everything in this paper about what is safer is a design argument backed by mechanism and by tests somebody can go and read, and not by data from the people it is for. A study with blind participants is the obvious next step, and it is also the thing most likely to overturn the premise, since users may well prefer a guess to an abstention, and the over-trigger rate we have been optimising against may not be what matters to them. We do not know.

Over-triggering, at 55% on text and 69.6% on speech, is the biggest open problem here. Rejection examples in the prompt, a second pass that classifies whether the request is in scope, and a confidence gate on the model [14] are all available and none of them is evaluated, because each would mean tuning against a frozen set, so a held-out set has to come first.

The spoken condition is two speakers, 59 records, 107 transcripts, one recogniser and one room. Neither speaker is a blind user and neither had used the system. The alignment gate drops the utterances the recogniser handled worst, so what we report is a floor on the degradation rather than an estimate of it. A bigger panel, more rooms and the Whisper path as a second recogniser would all sharpen it, though none of them is likely to reverse the direction, because the mechanism does not depend on how many speakers there are.

An earlier arm using a reasoning model, `qwen3:4b`, returned nothing parseable on 102 of 140 tier-1 calls at 6–8 s each, so tier 1 contributed nothing and the run scored exactly the keyword baseline. The validation boundary held and degraded to fewer capabilities rather than to wrong ones, which is what it is for, but that run says nothing about model size until thinking output is switched off.

The clock mapping is a camera-frame clock rather than an Orientation and Mobility one. Frame width covers 10 to 2 o'clock over roughly a 60° field of view, so “2 o'clock” means the right-hand edge of the frame and not the 90° a trained traveller would turn, and this needs relabelling or a real remapping before the system claims to fit O&M training.

Distance is coarse, roughly ±30–40% at 5 m with an uncalibrated focal constant, though the gating argument is about when to say a number rather than how good the number is, and a one-off per-device calibration would help any accuracy claim. The fabrication detector only flags invented objects, so it misses invented distances, bearings and counts, several of which turn up in the samples it did flag, and the number is a lower bound.

Conversational replies get counted and listed for inspection and excluded from the fabrication metric by definition, but we never measure whether they are right or well-calibrated, and they are part of what the system says now, so a user study would have to.

Finally, the remote-primary architecture assumes a laptop and a local network. §4 makes that failure safe rather than absent, and running on the phone alone waits on either a lighter detector head or hardware whose delegate partitions the model cleanly.

9 ETHICS, POSITIONALITY AND AVAILABILITY

The author is a sighted student developer and is not a member of the population this system is built for. None of it was co-designed with blind users and the one field walk was done by the author. We have tried to make that limitation do some work rather than just sit in a list, by tying every safety claim to a mechanism and a test that a reader can go and inspect, so that a study with blind participants could actually falsify the design arguments instead of merely failing to confirm them. No human-subjects data was collected and no ethics approval was needed for what is reported here, and a participant study is not attempted precisely because running one informally, with an unvalidated prototype and no approval, would be the wrong way to involve these users. The two speakers who recorded the spoken condition read a fixed list of system commands, took part voluntarily, and are identified only as A and B.

The system runs offline by design, with detection on a locally tethered laptop and speech recognition, synthesis and OCR on the handset, so no camera frame, no utterance and no location leaves the user's own devices, and the language model is local as well. The tether itself is an unencrypted local link, which is fine for a prototype on a personal hotspot and would need transport security before anybody deployed it.

Source code, the capability registry manifest, the frozen evaluation protocol, the 200-record labelled set and every run report are at github.com/Aditya17-bot/object_detection_blind. Clean-condition results carry the eval-set SHA-256 prefix `e4eeca83070e2d66`, and appending the spoken transcripts changes it to `f9e775b6a65279a4` with only the transcripts differing between the two. Any number quoted from a set with a different hash is not comparable.

10 CONCLUSION

For a user who cannot check what a system tells them, being able to decline is not the error path, it is a feature that has to be designed, built and measured at every layer where the thing can be wrong. We showed five of those in a working offline assistive system, each deciding on grounds that come from its own failure mode, and extended the idea to a voice agent whose language model gets to choose what the system does and never what it says about the world. The claim worth making is not that tool mediation prevents fabrication, which it plainly does, but that the same principle also explains a distance gate, a path threshold, a null-versus-empty distinction and a ceiling on how often the system speaks, three layers away from each other. The spoken-input result is what makes us believe it rather than just prefer it, since when the input got worse the mechanisms that abstain by construction were the ones still abstaining.

Tool	Argument	Backed by
walk	—	engine mode
find	class, required	find_message
describe	—	summarize_scene
count	class, required	count_message
recall	class, required	object memory
path	—	clear_path
check	direction, required	check_direction
read	—	on-device OCR
clock / zones	—	bearing style
sonar	on/off	sonar controller
mute	on/off, required	speech controller
stop / repeat	—	speech controller
abstain	template key	fixed template table

Table 4. The capability registry. One declarative table drives the recogniser's phrase list, the model's tool schema, the executor, and a manifest that both languages' test suites check against.

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